



CASE STUDY: STOCKDALE WASTEWATER LAGOON REMEDIATION

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A Cost-Effective Alternative to Mechanical Dredging

The City of Stockdale operates a three-lagoon wastewater system spanning nine acres and treating approximately 30,000 gallons per day. Built on clay bottoms and never dredged, the lagoons had accumulated more than 30 years of sludge, resulting in high BOD₅, TSS, ammonia, and periodic non-compliance with Texas Commission on Environmental Quality regulations. The condition was limiting future development.

Initial sonar (March 2025) confirmed average sludge depths of 5.6 feet in Lagoon #1, 1.3 feet in #2, and 2.0 feet in #3. Sludge was approximately 47 percent organic with compacted layers.

The Challenge

Aging Infrastructure and Excess Sludge

- Over 30 years of sludge accumulation, including material deposited during a 2013 flood
- Lagoon #1 sludge depths up to 5.8 feet
- Compacted sludge layers resistant to conventional biological digestion

Water Quality Concerns

Lagoon #1 effluent showed elevated levels of:

- Total Suspended Solids (TSS): 2,170 mg/L
- Ammonia (NH₃): 16.8 mg/L
- BOD₅: 65.3 mg/L

Financial Constraints

- Traditional dredging, hauling, and landfill disposal was projected to cost approximately \$10 million and would have required significant operational disruption.
- The City needed to restore compliance and capacity without shutting down the system.



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The E3 Solution

E3 implemented a structured two-phase remediation approach between July and December 2025.

Phase I, Assessment

Measured sludge at 18 points, confirmed organic composition, and validated in-situ treatment feasibility.

Phase II, Treatment

- Agitated compacted sludge using a Lagoon Crawler
- Deployed an NG 450 nanobubble generator across five distribution zones
- Dosed SRB MAX bacteria weekly at 2.5 gallons

Nanobubble oxygenation accelerated microbial digestion up to four to eight times faster than conventional aeration, eliminating the need for dredging.

Results

Sludge Reduction

- Lagoon #1 reduced from 5.8 ft to 1.0 ft (sludge judge) and 5.6 ft to 2.3 ft (sonar)
- Lagoons #2 and #3 reduced by approximately 50 percent

Effluent Improvements (Lagoon #1)

Parameter	Before	After	Reduction
TSS (mg/L)	2,170	120	94.5%
BOD ₅ (mg/L)	65.3	33	49.5%
NH ₃ (mg/L)	16.8	2.2	86.9%

Additional improvements included a **48 percent drop in organic sludge content** and significant visual water clarity gains.

Impact

- Avoided roughly two-thirds of projected dredging costs
- Eliminated an estimated 1,600 truckloads of sludge hauling
- Restored full system capacity
- Achieved full TCEQ compliance
- Enabled future city growth

This project demonstrates that nanobubble-enhanced bioremediation is a cost-effective, non-invasive alternative to mechanical dredging for municipal lagoon restoration.



Ask how Nano Gas can work for your community



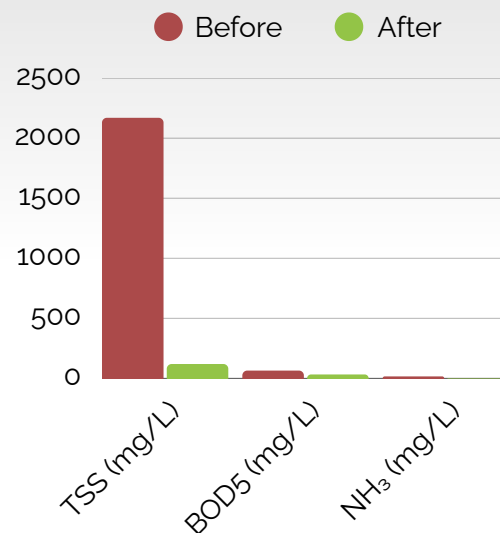
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- **TSS** measures how dirty or cloudy the water is from solids.
- **BOD₅** measures how much oxygen the water will consume because of organic waste.
- **NH₃** measures how much untreated nitrogen remains.